

# LIN7076 – Foundations of Computational Linguistics

## N-gram models

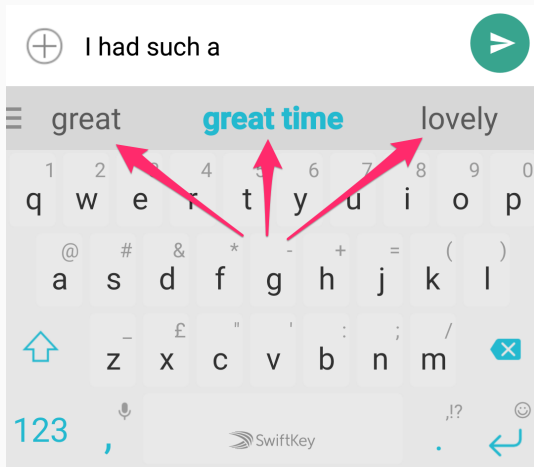
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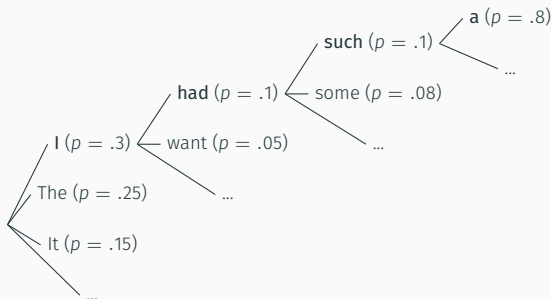
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# Generating text



# The intuition behind n-grams

- Your next word depends on the words you have used before.
- When predicting a word, pick the most likely, **given what was written before, within a specific window.**



- Likelihood (the  $ps$ ) gets approximated by computing **statistics over a similar-enough training corpus.**

# Plan for today

- Understand the  $n$ -gram **prediction process**.
- Understand how the probabilities used in that process are **approximated** from a training corpus.
- Understand how the resulting model can be **evaluated** against a test corpus.
- Explore **fixes/improvements** of shortcomings of the basic model.

## The n-gram generation process

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# Why n-gram?

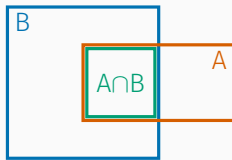
When predicting a word, pick the **most likely, given what was written before** within a specific window.

- We want to compute the conditional probability of a word  $w_i$ , given a bunch of preceding words  $[w_{i-n+1}, \dots, w_{i-1}]$ , that we call the **context**, of size  $n - 1$ .
- The sequence  $[w_{i-n+1}, \dots, w_{i-1}] + [w_i]$  is of size  $n$ . We call it a **n-gram**.
- For instance, the sequence  $[I, had, such, a]$ , is a 4-gram.
- A **n-gram model**, predicts the next word, given a  $(n - 1)$ -gram of preceding words.

# Refresher on conditional probabilities

- The **conditional probability** of an event  $A$ , given an event  $B$ , can be computed using the joint probability of  $A$  and  $B$ , and  $B$ 's probability.

$$P(A|B) = \frac{P(A, B)}{P(B)} = \frac{P(A \cap B)}{P(B)}$$



- Given any fixed event  $B$ , the probabilities  $\{P(X|B) \mid X\}$ , forms a probability distribution!

# Rephrasing the n-gram intuition using conditional probabilities

- Our target event  $A$ , is that of the word  $w_i$  being used. We call this word **target**.
- Our given event  $B$ , is that of the words  $[w_{i-n+1}, \dots, w_{i-1}]$  being used before. We call this sequence **context**.
- We want to compute  $P(A|B)$ , the probability of  $w_i$  given  $[w_{i-n+1}, \dots, w_{i-1}]$ !



# Rephrasing the n-gram intuition using conditional probabilities

- $P(w_i|[w_{i-n+1}, \dots, w_{i-1}])$  depends solely on the context, and the context appended with the target (= the  $n$ -gram!).

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$\begin{aligned} P(w_i|[w_{i-n+1}, \dots, w_{i-1}]) &= \frac{P([w_{i-n+1}, \dots, w_i])}{P([w_{i-n+1}, \dots, w_{i-1}])} \\ &= \frac{P(\text{context} + \text{target})}{P(\text{context})} \end{aligned}$$

$$P(\text{great}|I \text{ had such } a) = \frac{P(I \text{ had such } a \text{ great})}{P(I \text{ had such } a)}$$

- What would a **unigram model** look like?

## A 3-gram example with made-up probabilities

- Suppose our sentence starts with the bigram *I had*. Suppose that *such* and *some* are the only 2 possible continuations.

$$P(\text{such} | \text{I had}) = \frac{P(\text{I had such})}{P(\text{I had})} = .7$$

$$P(\text{some} | \text{I had}) = \frac{P(\text{I had some})}{P(\text{I had})} = .3$$

- Such* is more likely so we pick *such*. Now our sentence is *I had such*, and our context (bigram) becomes *had such*. Suppose that now *a* and *good* are the only 2 possible continuations.

$$P(a | \text{had such}) = \frac{P(\text{had such a})}{P(\text{had such})} = .8$$

$$P(\text{good} | \text{had such}) = \frac{P(\text{had such good})}{P(\text{had such})} = .2$$

- A* is more likely, so we pick *a*. Now our sentence is *I had such a*, and our context becomes *such a*. Etc...

## A potential issue with fixed window sizes

When predicting a word, pick the most likely, given what was written before **within a specific window**.

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<sup>1</sup>You can try to do the same with only relative clauses!

## A potential issue with fixed window sizes

When predicting a word, pick the most likely, given what was written before **within a specific window**.

- Why use a window, especially a short one? Natural language is recursive after all!
- Consider (1), which features **center-embedding**, in the form of an arbitrary number of nested complementizer phrases, between the main subject and its verb..<sup>1</sup>

(2) The *student* who Jo said that Ed thinks that ... Al likes \_\_ ...  
subject arbitrarily long, nested relative clauses

- Could a  $n$ -gram model with a fixed window size correctly predict the number marking on the following verb? Could it even predict that it should be a verb?

<sup>1</sup>You can try to do the same with only relative clauses!

## Why windows, anyway

- Using a fixed window makes the assumption that most of the time, language generation is oblivious of what was said too long ago. As we've just seen, this is not always true.
- This is known as the **Markov assumption**, which applies to processes well beyond CompLing! Biology, economy, robotics...
- Even if imperfect, the Markov assumption is computationally more efficient, as it requires us to compute  $n$ -gram and  $n - 1$ -gram **occurrences** for a fixed  $n$ , as we will soon see.
- It will also prevent the model from “memorizing” big chunks of the training data, something known as overfitting.
- Can you think of what shorter/longer windows would on average best capture in terms of syntax/semantics?

## A note on sampling

When predicting a word, **pick the most likely**, given what was written before within a specific window.

- What if two possible targets  $w_i$  and  $w'_i$  are associated to very close conditional probabilities, e.g. .999 and .1? Should we still pick the most likely, all the time?

# A note on sampling

When predicting a word, **pick the most likely**, given what was written before within a specific window.

- What if two possible targets  $w_i$  and  $w'_i$  are associated to very close conditional probabilities, e.g. .999 and .1? Should we still pick the most likely, all the time?
- In practice, we **sample**: we pick target words randomly, giving more chances to the more likely ones.
- One simple way to think about it: drawing one ball from a big lottery box such that each target word is assigned a ball color, and gets as many balls as its conditional probability converted to a round percentage (so  $p = .155$  means 16 balls).



# Sampling example

- Remember our made up conditional probabilities for *such* and *some*, given the context *I had*?

$$P(\text{such} | I \text{ had}) = \frac{P(I \text{ had such})}{P(I \text{ had})} = .7$$
$$P(\text{some} | I \text{ had}) = \frac{P(I \text{ had some})}{P(I \text{ had})} = .3$$

- Instead of returning *such* (presumed most likely) 100% of the time, sampling will return *such* 70% of the time, and *some* 30% of the time.
- More generally, any target word *w* would be picked  $(100 \times P(w | I \text{ had}))\%$  of the time.



## Approximating conditional probabilities

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# From probabilities to counts

- So far we have reasoned about hypothetical probabilities of words or sequences of words. **How are these probabilities estimated?**
- The probability of a sequence of words  $S$  of any fixed length (can be 1, i.e.  $S$  is a single word) can be estimated based on a training corpus.
- This is done by counting the number of  $S$ 's occurrences, and dividing it up by the sum of the counts of all same-length sequences.
- The [Google  \$n\$ -gram Viewer](#) actually does this on a very large corpus of books!

## From probabilities to counts

$$P(S) = \frac{C(S)}{\sum_{|S'|=|S|} C(S')} \triangleq \alpha C(S)$$

- Suppose now  $S'$  is obtained from  $S$  by removing  $S$ 's last word:  $S = S' + w$ . If  $S$  is a  $n$ -gram,  $S'$  is a  $(n - 1)$ -gram.  $S$ 's probability can be expressed by summing the probabilities of its immediate continuations...

$$P(S') = \sum_w P(S' + w) = \sum_w \alpha C(S' + w) = \alpha \sum_w C(S' + w) = \alpha C(S')$$

- We have just expressed probabilities of  $n$  and  $(n - 1)$ -grams using their **counts** and the **same normalizing factor**  $\alpha$ !

## From conditional probabilities to counts

$$\begin{aligned}P([w_{i-n+1}, \dots, w_i]) &= \alpha C([w_{i-n+1}, \dots, w_i]) \\P([w_{i-n+1}, \dots, w_{i-1}]) &= \alpha C([w_{i-n+1}, \dots, w_{i-1}])\end{aligned}$$

- Plugging this into our  $n$ -gram prediction formula, the  $\alpha$ s eventually cancel out! So we get a conditional probability that solely depends on  $n$ -gram counts.

$$\begin{aligned}P(w_i | [w_{i-n+1}, \dots, w_{i-1}]) &= \frac{P([w_{i-n+1}, \dots, w_i])}{P([w_{i-n+1}, \dots, w_{i-1}])} \\&\approx \frac{C([w_{i-n+1}, \dots, w_i])}{C([w_{i-n+1}, \dots, w_{i-1}])} \\&\approx \frac{C(\text{context} + \text{target})}{C(\text{context})}\end{aligned}$$

# Dealing with sentence boundaries

$$P(w_i | [w_{i-n+1}, \dots, w_{i-1}]) \simeq \frac{C([w_{i-n+1}, \dots, w_i])}{C([w_{i-n+1}, \dots, w_{i-1}])} = \frac{C(\text{context} + \text{target})}{C(\text{context})}$$

- The above equation predicts the next word, given a context of size  $n - 1$ . But what should we do at the very beginning of a sentence, when the context is empty? And when should we stop?
- To properly initiate and terminate generation, we posit two extra silent “words”: **<s>** (“**beginning** of sentence”) and **</s>** (“**end** of sentence”).
- For a  $n$ -gram model, each sentence in the training corpus gets **padded with  $n - 1$  <s> at the beginning, and 1 </s> at the end.** Counts incorporate these extra words.
- To start generating a sentence, we can then assume an initial, silent context made of  $n - 1$  <s> symbols!
- And we stop as soon as </s> gets generated.

## Estimating conditional probabilities: a worked example

- Let's compute a toy bigram model. We pad with 1 `<s>` symbol.

- (3)
- a. `<s>` my dog is nice `</s>`
  - b. `<s>` my cat is nasty `</s>`
  - c. `<s>` your cat hates my cat `</s>`

## Estimating conditional probabilities: a worked example

- Let's compute a toy bigram model. We pad with 1 `<s>` symbol.

(4) a. `<s>` my dog is nice `</s>`

b. `<s>` my cat is nasty `</s>`

c. `<s>` your cat hates my cat `</s>`

- Bigrams: (`<s>`, my), (my, dog), (dog, is), (is, nice), (nice, `</s>`), (`<s>`, my), (my, cat), (cat, is), (is, nasty), (nasty, `</s>`), (`<s>`, Your), (Your, cat), (cat, hates), (hates, my), (my, cat), (cat, `</s>`).
- Unigrams: `<s>`, my, dog, is, nice, `</s>`, `<s>`, my, cat, is, nasty, `</s>`, `<s>`, Your, cat, hates, my, cat, `</s>`.

## Estimating conditional probabilities: a worked example

- (5) a.  $\langle s \rangle$  my dog is nice  $\langle /s \rangle$   
b.  $\langle s \rangle$  my cat is nasty  $\langle /s \rangle$   
c.  $\langle s \rangle$  your cat hates my cat  $\langle /s \rangle$

| $w_i \rightarrow$<br>$\downarrow w_{i-1}$ | my | dog | is | nice | cat | nasty | your | hates | $\langle /s \rangle$ | Total |
|-------------------------------------------|----|-----|----|------|-----|-------|------|-------|----------------------|-------|
| $\langle s \rangle$                       | 2  | 0   | 0  | 0    | 0   | 0     | 1    | 0     | 0                    | 3     |
| my                                        | 0  | 1   | 0  | 0    | 2   | 0     | 0    | 0     | 0                    | 3     |
| dog                                       | 0  | 0   | 1  | 0    | 0   | 0     | 0    | 0     | 0                    | 1     |
| is                                        | 0  | 0   | 0  | 1    | 0   | 1     | 0    | 0     | 0                    | 2     |
| nice                                      | 0  | 0   | 0  | 0    | 0   | 0     | 0    | 0     | 1                    | 1     |
| cat                                       | 0  | 0   | 1  | 0    | 0   | 0     | 0    | 1     | 1                    | 3     |
| nasty                                     | 0  | 0   | 0  | 0    | 0   | 0     | 0    | 0     | 1                    | 1     |
| your                                      | 0  | 0   | 0  | 0    | 1   | 0     | 0    | 0     | 0                    | 1     |
| hates                                     | 1  | 0   | 0  | 0    | 0   | 0     | 0    | 0     | 0                    | 1     |
| Total                                     | 3  | 1   | 2  | 1    | 3   | 1     | 1    | 1     | 3                    |       |



$$P(\text{my}|\text{<s>}) = \frac{C(\text{<s>my})}{C(\text{<s>})} = \frac{2}{3}$$

| $w_i \rightarrow$<br>$\downarrow w_{i-1}$ | my | dog | is | nice | cat | nasty | your | hates | </s> | Total |
|-------------------------------------------|----|-----|----|------|-----|-------|------|-------|------|-------|
| <s>                                       | 2  | 0   | 0  | 0    | 0   | 0     | 1    | 0     | 0    | 3     |
| my                                        | 0  | 1   | 0  | 0    | 2   | 0     | 0    | 0     | 0    | 3     |
| dog                                       | 0  | 0   | 1  | 0    | 0   | 0     | 0    | 0     | 0    | 1     |
| is                                        | 0  | 0   | 0  | 1    | 0   | 1     | 0    | 0     | 0    | 2     |
| nice                                      | 0  | 0   | 0  | 0    | 0   | 0     | 0    | 0     | 1    | 1     |
| cat                                       | 0  | 0   | 1  | 0    | 0   | 0     | 0    | 1     | 1    | 3     |
| nasty                                     | 0  | 0   | 0  | 0    | 0   | 0     | 0    | 0     | 1    | 1     |
| your                                      | 0  | 0   | 0  | 0    | 1   | 0     | 0    | 0     | 0    | 1     |
| hates                                     | 1  | 0   | 0  | 0    | 0   | 0     | 0    | 0     | 0    | 1     |
| Total                                     | 3  | 1   | 2  | 1    | 3   | 1     | 1    | 1     | 3    |       |

$$P(\text{dog}|\text{my}) = \frac{C(\text{my dog})}{C(\text{my})} = \frac{1}{3}$$

| $w_i \rightarrow$<br>$\downarrow w_{i-1}$ | my | dog | is | nice | cat | nasty | your | hates | </s> | Total |
|-------------------------------------------|----|-----|----|------|-----|-------|------|-------|------|-------|
| <s>                                       | 2  | 0   | 0  | 0    | 0   | 0     | 1    | 0     | 0    | 3     |
| my                                        | 0  | 1   | 0  | 0    | 2   | 0     | 0    | 0     | 0    | 3     |
| dog                                       | 0  | 0   | 1  | 0    | 0   | 0     | 0    | 0     | 0    | 1     |
| is                                        | 0  | 0   | 0  | 1    | 0   | 1     | 0    | 0     | 0    | 2     |
| nice                                      | 0  | 0   | 0  | 0    | 0   | 0     | 0    | 0     | 1    | 1     |
| cat                                       | 0  | 0   | 1  | 0    | 0   | 0     | 0    | 1     | 1    | 3     |
| nasty                                     | 0  | 0   | 0  | 0    | 0   | 0     | 0    | 0     | 1    | 1     |
| your                                      | 0  | 0   | 0  | 0    | 1   | 0     | 0    | 0     | 0    | 1     |
| hates                                     | 1  | 0   | 0  | 0    | 0   | 0     | 0    | 0     | 0    | 1     |
| Total                                     | 3  | 1   | 2  | 1    | 3   | 1     | 1    | 1     | 3    |       |

$$P(\text{hates}|\text{cat}) = \frac{C(\text{cat hates})}{C(\text{cat})} = \frac{1}{3}$$

| $w_i \rightarrow$<br>$\downarrow w_{i-1}$ | my | dog | is | nice | cat | nasty | your | hates | </s> | Total |
|-------------------------------------------|----|-----|----|------|-----|-------|------|-------|------|-------|
| <s>                                       | 2  | 0   | 0  | 0    | 0   | 0     | 1    | 0     | 0    | 3     |
| my                                        | 0  | 1   | 0  | 0    | 2   | 0     | 0    | 0     | 0    | 3     |
| dog                                       | 0  | 0   | 1  | 0    | 0   | 0     | 0    | 0     | 0    | 1     |
| is                                        | 0  | 0   | 0  | 1    | 0   | 1     | 0    | 0     | 0    | 2     |
| nice                                      | 0  | 0   | 0  | 0    | 0   | 0     | 0    | 0     | 1    | 1     |
| cat                                       | 0  | 0   | 1  | 0    | 0   | 0     | 0    | 1     | 1    | 3     |
| nasty                                     | 0  | 0   | 0  | 0    | 0   | 0     | 0    | 0     | 1    | 1     |
| your                                      | 0  | 0   | 0  | 0    | 1   | 0     | 0    | 0     | 0    | 1     |
| hates                                     | 1  | 0   | 0  | 0    | 0   | 0     | 0    | 0     | 0    | -     |
| Total                                     | 3  | 1   | 2  | 1    | 3   | 1     | 1    | 1     | 3    |       |

- Can you guess which sentence(s) would be most likely to be generated by this bigram model?

## Evaluating n-gram models

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# Probability of a corpus

- To evaluate how good a  $n$ -gram model is, one needs new data: typically a test corpus that is not too different from the training corpus.<sup>2</sup>
- We can then evaluate how likely this whole corpus is to be generated by our  $n$ -gram model!
- This is thanks to the chain rule:

$$\begin{aligned} P(A_1 \cap A_2 \cap \dots \cap A_k) &= P(A_1) \times \prod_{i=2}^k P(A_i | A_1 \cap \dots \cap A_{i-1}) \\ &\stackrel{\text{Markov}}{=} P(A_1) \times \prod_{i=2}^k P(A_i | A_{i-n+1} \cap \dots \cap A_{i-1}) \\ P([w_1 \dots w_k]) &\stackrel{\text{Markov}}{=} P(w_1) \times \prod_{i=2}^k P(w_i | [w_{n-i+1} \dots w_{i-1}]) \end{aligned}$$

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<sup>2</sup>Why not test on the training corpus?

# From probability to perplexity

- Measuring the performance of a  $n$ -gram based on the probability of a test corpus is nice, but depends on the corpus size: bigger corpora will mechanically be less probable (why?).
- To avoid this issue, we use a normalized variant of corpus probability called **perplexity**. The lower the perplexity, the better the model.

$$\begin{aligned} PPL([w_1 \dots w_k]) &= P([w_1 \dots w_k])^{-\frac{1}{k}} \\ &\stackrel{\text{Markov}}{=} \left( P(w_1) \times \prod_{i=2}^k P(w_i | [w_{n-i+1} \dots w_{i-1}]) \right)^{-\frac{1}{k}} \end{aligned}$$

- If  $k \uparrow$ ,  $\frac{1}{k} \downarrow$ ,  $-\frac{1}{k} \uparrow$  and  $p^{-\frac{1}{k}} \downarrow$  (with  $p \geq 0$ ). So if two corpora have same probability, the larger one will get a smaller perplexity.

## An issue with perplexity

$$PPL([w_1 \dots w_k]) \stackrel{\text{Markov}}{=} \left( P(w_1) \times \prod_{i=2}^k P(w_i | [w_{n-i+1} \dots w_{i-1}]) \right)^{-\frac{1}{k}}$$

- Looking at the above product of probabilities, we notice that if only one probability is 0, the whole product is  $\frac{1}{0} = +\infty$ !
- This would happen if the test corpus contains **one  $n$ -gram that the training corpus does not have** – which is far from unlikely!
- So perplexity is **unlikely to distinguish good from bad  $n$ -gram models**, as we defined them. Both kinds will likely get an infinite perplexity.
- More generally, probabilities learned from a finite training corpus will necessarily be 0 for many perfectly reasonable  $n$ -grams...bringing us to the **competence-performance distinction**.

# A solution: smoothing

- To prevent perplexity from vanishing, we can simply **add a small increment** to all our  $n$ -gram counts!

| $w_i \rightarrow$<br>$\downarrow w_{i-1}$ | <s>        | my             | dog            | is             | nice           | cat            | nasty          | your           | hates          | </s>           | Total          |
|-------------------------------------------|------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| <s>                                       | $\epsilon$ | $2+\epsilon$   | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $1+\epsilon$   | $\epsilon$     | $\epsilon$     | $3+10\epsilon$ |
| my                                        | $\epsilon$ | $\epsilon$     | $1+\epsilon$   | $\epsilon$     | $\epsilon$     | $2+\epsilon$   | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $3+10\epsilon$ |
| dog                                       | $\epsilon$ | $\epsilon$     | $\epsilon$     | $1+\epsilon$   | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $1+10\epsilon$ |
| is                                        | $\epsilon$ | $\epsilon$     | $\epsilon$     | $\epsilon$     | $1+\epsilon$   | $\epsilon$     | $1+\epsilon$   | $\epsilon$     | $\epsilon$     | $\epsilon$     | $2+10\epsilon$ |
| nice                                      | $\epsilon$ | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $1+\epsilon$   | $1+10\epsilon$ |
| cat                                       | $\epsilon$ | $\epsilon$     | $\epsilon$     | $1+\epsilon$   | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $1+\epsilon$   | $1+\epsilon$   | $3+10\epsilon$ |
| nasty                                     | $\epsilon$ | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $1+\epsilon$   | $1+10\epsilon$ |
| your                                      | $\epsilon$ | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $1+\epsilon$   | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $1+10\epsilon$ |
| hates                                     | $\epsilon$ | $1+\epsilon$   | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $1+10\epsilon$ |
| </s>                                      | $\epsilon$ | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | $\epsilon$     | -              |
| Total                                     | -          | $3+10\epsilon$ | $1+10\epsilon$ | $2+10\epsilon$ | $1+10\epsilon$ | $3+10\epsilon$ | $1+10\epsilon$ | $1+10\epsilon$ | $1+10\epsilon$ | $3+10\epsilon$ |                |

- How do you think increasing/decreasing  $\epsilon$  will affect training set perplexity? Test set perplexity?



# How smoothing affects conditional probabilities

$$\begin{aligned} P(w_i | [w_{i-n+1}, \dots, w_{i-1}]) &\simeq \frac{C_{\text{smooth}}([w_{i-n+1}, \dots, w_i])}{C_{\text{smooth}}([w_{i-n+1}, \dots, w_{i-1}])} \\ &\simeq \frac{C([w_{i-n+1}, \dots, w_i]) + \epsilon}{C([w_{i-n+1}, \dots, w_{i-1}]) + \epsilon|L|} \end{aligned}$$

- With  $|L|$  the size of the lexicon (=set of unigrams).
- Why do we get  $\epsilon|L|$  in the denominator and not just  $\epsilon$ ?
- Recall that the number of occurrences of a  $(n-1)$ -gram can be expressed as the number of occurrences of  $n$ -gram continuations.

$$\begin{aligned} C_{\text{smooth}}([w_{i-n+1}, \dots, w_{i-1}]) &= \sum_{w \in L} C_{\text{smooth}}([w_{i-n+1}, \dots, w_{i-1}, w]) \\ &= \sum_{w \in L} (C([w_{i-n+1}, \dots, w_{i-1}, w]) + \epsilon) \\ &= \sum_{w \in L} C([w_{i-n+1}, \dots, w_{i-1}, w]) + \epsilon|L| \\ &= C([w_{i-n+1}, \dots, w_{i-1}]) + \epsilon|L| \end{aligned}$$

## Another solution: stupid backoff

- Instead of messing with all the  $n$ -gram counts, one can revert to a  $(n - 1)$ -gram model, just in case the  $n$ -gram probability is 0.
- If for instance  $P(\text{great}|\text{I had such a})$  is 0, then maybe  $P(\text{great}|\text{had such a})$  is not? And if it is 0 too, then let's try  $P(\text{great}|\text{such a})$ ! Etc.
- This kind of hybrid model can be expressed in the following, recursive<sup>3</sup> way ( $\lambda$  being some discounting factor).

$$SB(w_i|[w_{i-n+1}, \dots, w_{i-1}]) = \begin{cases} P(w_i|[w_{i-n+1}, \dots, w_{i-1}]) & \text{if } > 0 \\ \lambda SB(w_i|[w_{i-n+2}, \dots, w_{i-1}]) & \text{otherwise} \end{cases}$$

- This model is no longer a proper probability distribution. Additionally, it requires us to compute  $k$ -gram occurrences for potentially every  $k < n$ !
- But unlike the smoothing solution, it may assign different probabilities to unlikely vs. super unlikely sequences.

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<sup>3</sup>Notice how the “otherwise” case calls SB on a smaller context.

<https://colab.research.google.com/drive/1tCpAVF-SWnosVbRek8gpD9iz1g8lRhiNo>